HR Workforce Analytics Dashboard — Power BI

An end-to-end Power BI project analyzing a 1,000-employee HR dataset to answer real workforce planning questions: diversity representation, geographic footprint, and pay equity across roles and departments. Built as a personal project to apply and demonstrate skills.

The Business Problem

HR and People Operations teams are regularly asked questions like "how diverse is our workforce?", "where are we geographically concentrated?" and "is pay consistent across roles and departments?"; but without a centralized, trustworthy report, these questions usually get answered with slow, manual Excel pulls that go hard immediately and are error-prone to rebuild each time.

This dashboard solves that by turning a raw, messy HR export into a governed, reusable data model with a single source of truth; so, any of these questions can be answered in seconds, self-served, and refreshed automatically as new data arrives, instead of requiring a fresh manual analysis every time leadership asks.

Dataset

Real-world style HR dataset (1,000 employee records, 17 attributes) sourced from FP20Analytics Power BI Data Challenge #15. Fields include demographics (gender, ethnicity, age), employment details (hire/exit date, job title, department, business unit), compensation (annual salary, bonus %), and location (country, city, latitude/longitude).

Tools & Skills Used

- **Power Query (M)** — data cleaning, staging pattern, star schema construction via reference queries and merges

- **Power BI Data Modeling** — star schema design with surrogate keys, role-playing date dimension

- **DAX** — time intelligence, dynamic point-in-time calculations, iterative row-context filtering

- **Power BI Desktop / Report Design** — custom visual layout, conditional formatting, UI/UX driven dashboard design

1. Data Cleaning (Power Query)

Key transformations applied before modeling:

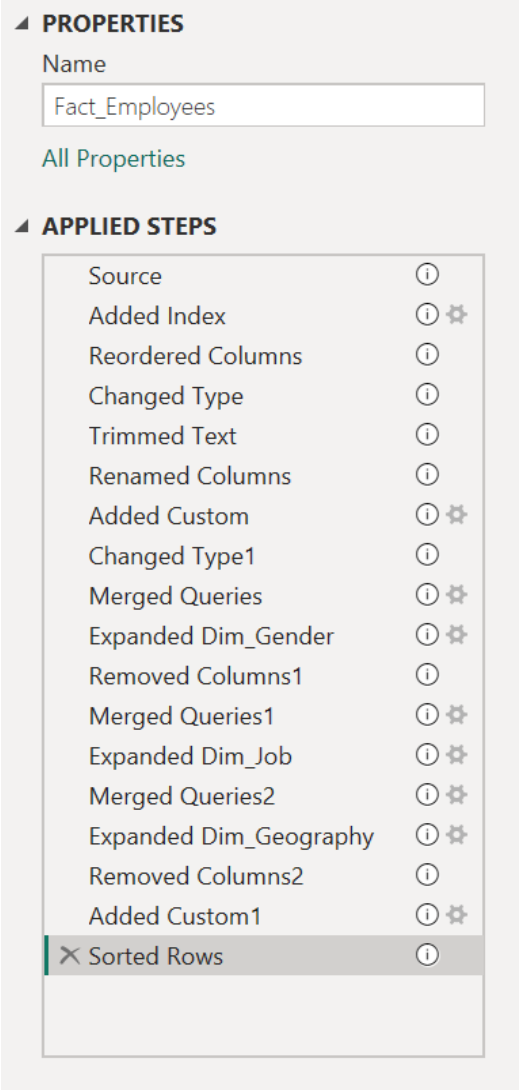
- Data type correction across all columns (dates, percentages, whole numbers), auto-detection isn't reliable and silently breaks DAX aggregations if left unchecked.

- Text cleaning (Trim) on all categorical text fields to eliminate invisible whitespace that would otherwise fragment slicers and groupings.

- Data quality investigation: discovered that “Employee ID” is not a unique key — 172 records share an Employee ID with a completely different person. Identified “Employee New ID” as the true unique identifier and used it throughout the model instead.

- Derived column: “Employment Status” (Active / Exited) based on whether ‘Exit Date’ is populated — computed once during ETL rather than repeated inside every DAX measure.

- Staging query pattern: the cleaned employee table was frozen as a non-loading staging query (‘Stg_Employees’, Enable Load disabled), with the final Fact and Dimension tables built as independent references off that frozen point, avoiding the common Power Query trap where editing an upstream query after other queries reference it silently breaks them.



2. Data Modeling (Star Schema)

Built a proper star schema instead of one flat table:

| Table | Type | Contents |

| ‘Fact_Employees’ | Fact | Employee-level records: salary, bonus %, hire/exit date, age, employment status, and foreign keys to each dimension |

| ‘Dim_Job’ | Dimension | Job Title, Department, Business Unit (245 unique combinations) |

| ‘Dim_Geography’ | Dimension | Country, City, Latitude, Longitude (13 unique locations) |

| ‘Dim_Demographics’ | Dimension | Gender, Ethnicity (8 unique combinations) |

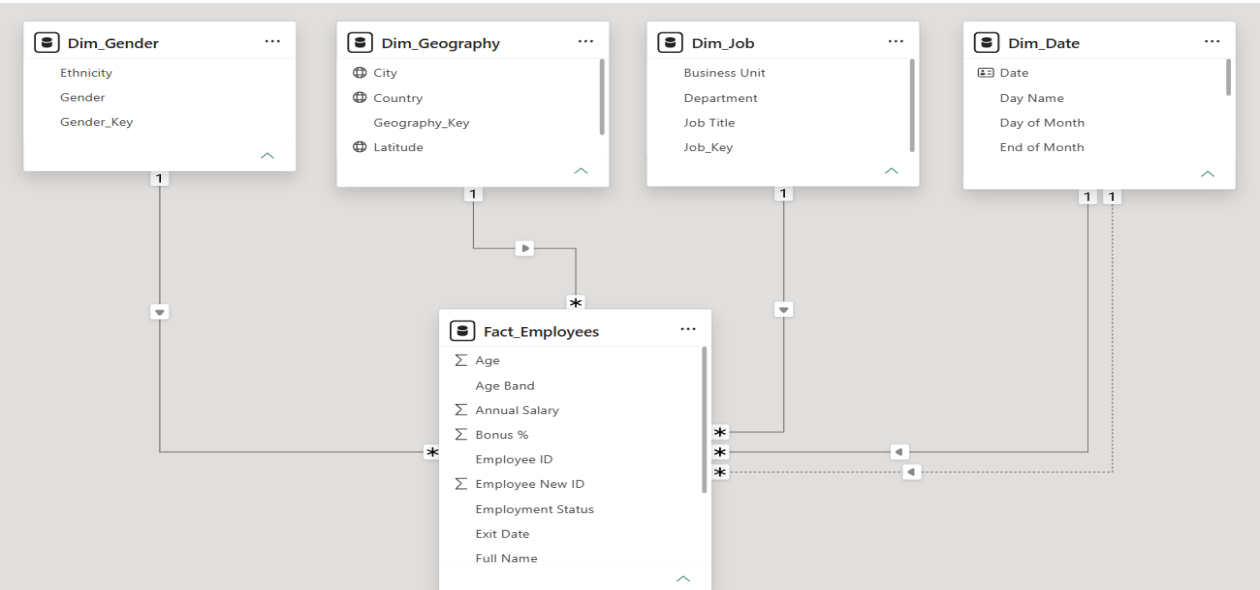
| ‘Dim_Date’ | Dimension | Full calendar table (1992–2022), marked as an official Date Table |

Design decisions worth noting:

- Each dimension uses a proper surrogate integer key (Kimball-style), generated via Power Query's Index Column and resolved onto the fact table through Merge Queries — rather than natural/business keys, which is more resilient to future source data changes.

- ‘Job Title’ alone is not a unique key (e.g., "Director" spans 7 different departments) — this was verified before modeling to avoid building an incorrect relationship.

- ‘Dim_Date’ has two relationships to the fact table: an active one on ‘Hire Date’ and an inactive one on ‘Exit Date’ (activated selectively in DAX via ‘USERELATIONSHIP’ where needed), since a single date table can't be simultaneously active against two different fact-table date columns without ambiguity.



3. Key DAX Measures

Created a separate Measures table and giving below the few measures' formula:

Total Employees = DISTINCTCOUNT (Fact_Employees[Employee New ID])

Active Employees = CALCULATE ([Total Employees], Fact_Employees[Employment Status] = "Active")

% of Total Employees = DIVIDE ([Total Employees], CALCULATE ([Total Employees], ALL (Fact_Employees)))

Avg Annual Salary = AVERAGE (Fact_Employees[Annual Salary])

4. Report Pages

Page 1 — Workforce Diversity: Gender, ethnicity, and age-band distribution, plus an ethnicity × gender matrix with conditional formatting to surface intersectional patterns that single-dimension charts can't show.

Finding: Black employees represent just 7.4% of the workforce — notably lower than the other three ethnic groups — while gender balance stays close to even (~51/49) within every group.

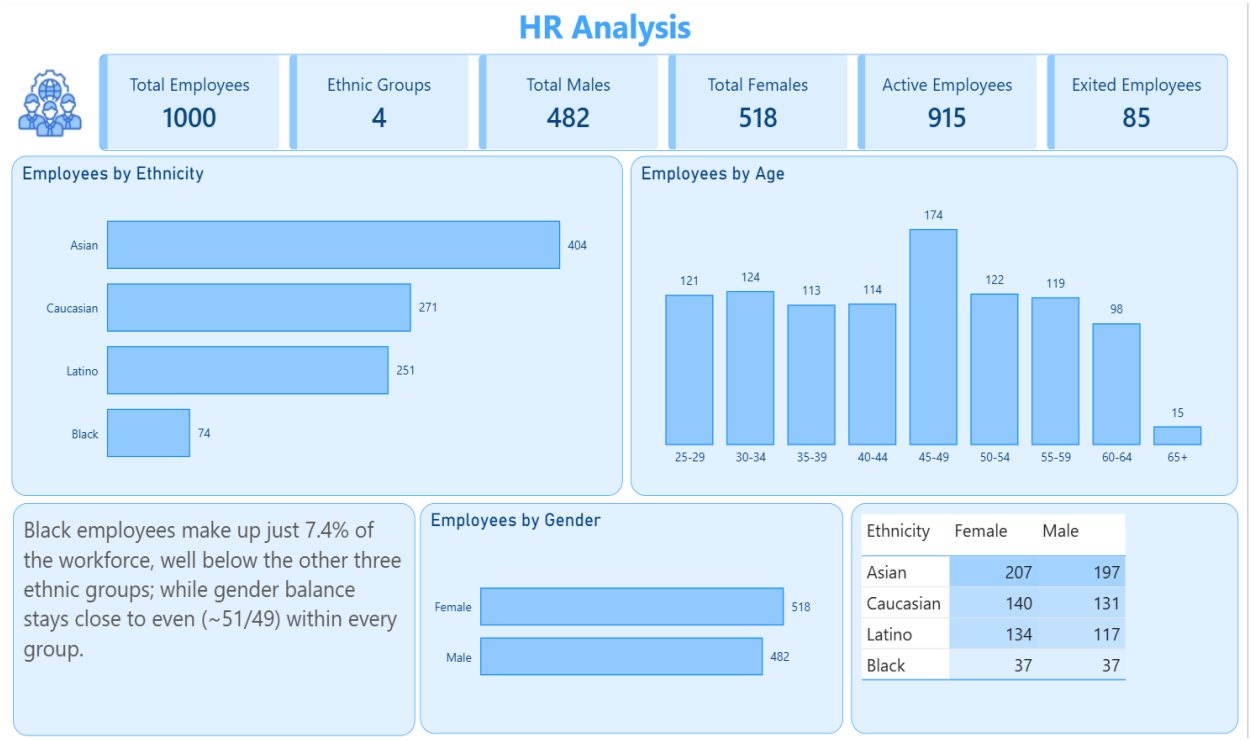
Page 2 — Geographic Distribution: Interactive bubble map (using true latitude/longitude coordinates, not text-based auto-geocoding, for accuracy) plus a tree-map for precise headcount comparison across offices.

Finding: 64% of the workforce is US-based across 6 cities, with Seattle as the largest single office (118 employees). Despite the geographic spread, average salary is nearly identical across all three countries (~$112K–114K) — no evidence of location-based pay disparity.

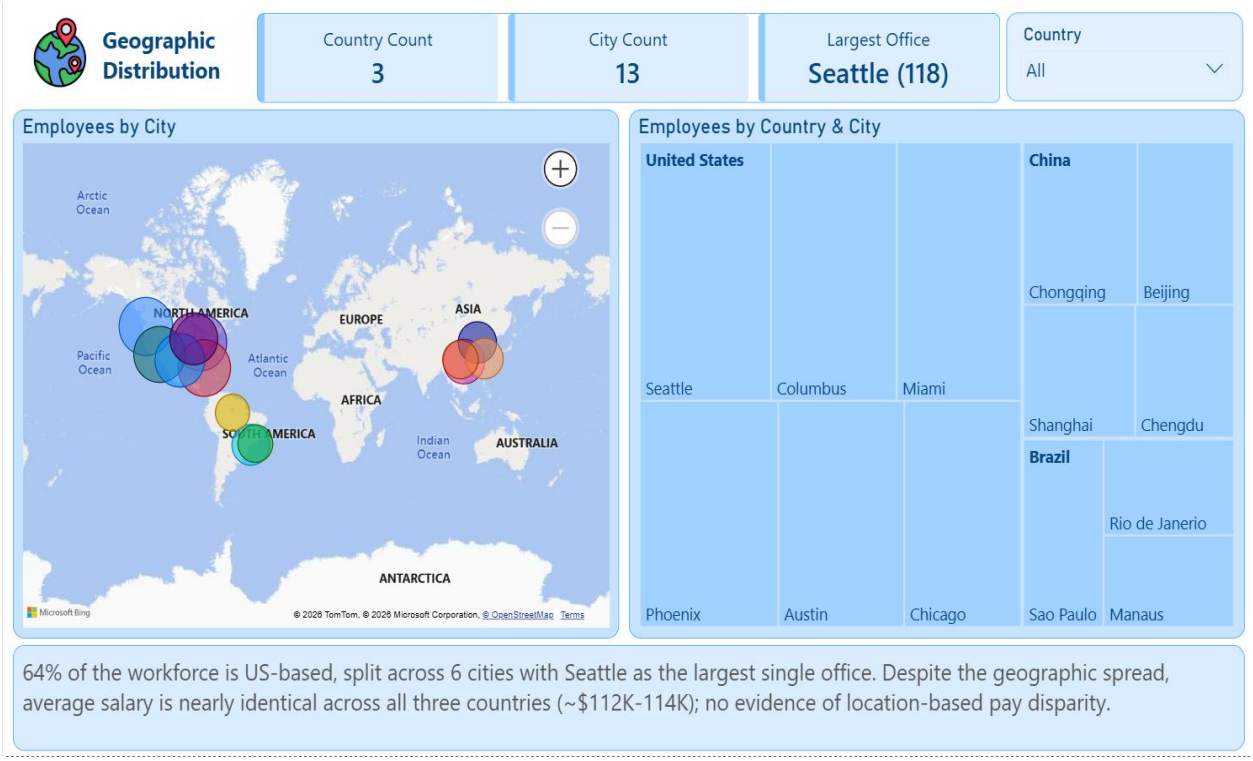
Page 3 — Pay Analysis: Average salary by job title and by department, with the 3-person Board Member/executive tier deliberately isolated from comparative visuals (their $258K average pay is roughly double the next tier and would distort any shared color/axis scale if included).

Finding: Excluding the executive tier, department averages span a fairly tight $98K–$129K range. Marketing pays the most among individual-contributor departments ($129K), while IT trails every other department at $98K — a pattern that holds consistently across the ethnicity/gender and business-unit breakdowns elsewhere in the analysis, not just as an isolated department-level average. Job Title tells a complementary story: System Administrator ($203K) unusually outpaces Director ($172K), which would be worth validating with the business in a real engagement rather than assuming it's correct at face value.

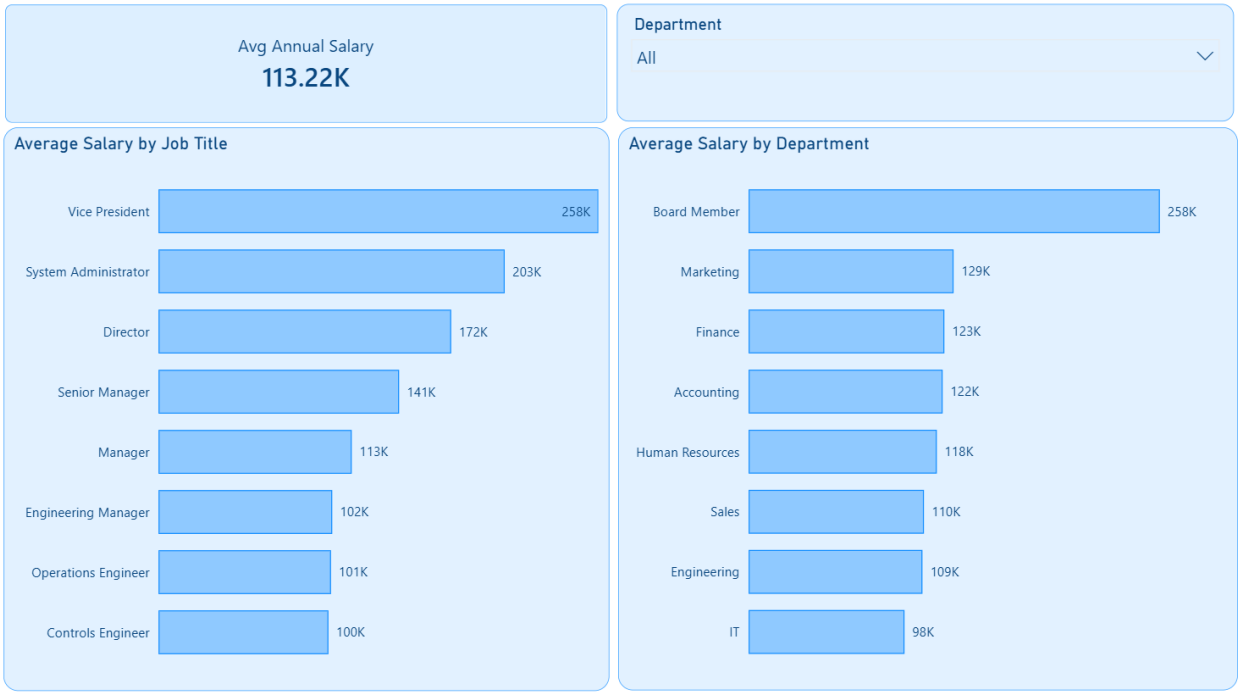
Page-1: Workforce Diversity



Page-2: Geographic Distribution



Page-3: Salary & Department



What I'd Build Next

With more time, this model is already structured to support: turnover rate trend (using the same `Average Number of Employees` cohort logic as Retention Rate), pay equity analysis controlling for job title and tenure simultaneously, and a bonus-trend page using the existing 'Dim_Date' relationship.

About This Project

Built by Momna Sajid as a self-directed portfolio project including data quality investigation, star schema design, and defensible handling of outliers and partial-year data.